Application E1 - Problem Statement:

A car of mass M moves on a rough horizontal road at a velocity $\vec{V_0}$. At $t_0 = 0$, the driver brakes, and the car stops at time t_1. The frictional forces are represented by constant forces $\vec{f}$ and $\vec{f'}$, both parallel to the road and opposite to the movement.

a. Express t_1 in terms of f, f', M, and V_0. b. Express the traveled distance D in terms of M, V_0, f, and f'.

Part a: Finding the stopping time (t_1)

For this part, we will use the **Theorem of Linear Momentum** (also known as the Impulse-Momentum Theorem), as it relates force and time.

1. Define the System and States:

- System: The car of mass M.
- Initial State (at $t_0 = 0$):
  - Initial Velocity: $\vec{V_0}$
  - Initial Momentum: $\vec{P_0} = M\vec{V_0}$
- Final State (at t_1):
  - Final Velocity: $\vec{V_f} = \vec{0}$ (the car stops)
  - Final Momentum: $\vec{P_f} = \vec{0}$
- Time Interval: $\Delta t = t_1 - t_0 = t_1$

2. Analyze the External Forces ($\Sigma\vec{F}_{ext}$):

- Vertical Forces:
  - Weight ($\vec{W}$ or $M\vec{g}$): Acts downwards.
  - Normal Force ($\vec{N}$): Acts upwards from the road.
  - These forces cancel each other out on the horizontal road: $\vec{W} + \vec{N} = \vec{0}$.
- Horizontal Forces (Opposing Motion):
  - Friction $\vec{f}$
  - Braking force $\vec{f'}$
  - The net external force is $\Sigma\vec{F}_{ext} = \vec{W} + \vec{N} + \vec{f} + \vec{f'} = \vec{f} + \vec{f'}$.

3. Apply the Linear Momentum Theorem: From your notes (Section D, case b), for a non-isolated system with a constant external force:

$$\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$$

$$\vec{P_f} - \vec{P_0} = (\vec{f} + \vec{f'}) \cdot t_1$$

4. Solve using Projections: Let's define the positive x-axis ($\vec{i}$) as the direction of the initial velocity $\vec{V_0}$.

- $\vec{V_0} = V_0\vec{i} \implies \vec{P_0} = MV_0\vec{i}$

- $\vec{P}_f = \vec{0}$
- Since $\vec{f}$ and $\vec{f'}$ oppose the motion, they point in the negative x-direction:
  - $\vec{f} = -f\vec{i}$
  - $\vec{f'} = -f'\vec{i}$

Now, substitute these vectors into the equation:

$$\vec{0} - (MV_0\vec{i}) = (-f\vec{i} - f'\vec{i}) \cdot t_1$$

$$-MV_0\vec{i} = -(f + f')\vec{i} \cdot t_1$$

Looking at the scalar components:

$$-MV_0 = -(f + f') \cdot t_1$$

Solving for t_1:

$$t_1 = \frac{-MV_0}{-(f + f')}$$

$$t_1 = \frac{MV_0}{f + f'}$$

Part b: Finding the stopping distance (D)

For this part, which relates force and distance, we use the **non-conservation of mechanical energy equation**. The work done by the non-conservative friction forces equals the change in the system's mechanical energy.

1. Define the States for Energy:

- **Initial State (at $x = 0$):**
  - Velocity: V_0
  - Kinetic Energy: $KE_0 = \frac{1}{2}MV_0^2$
  - Potential Energy: $GPE_0 = 0$ (we set the horizontal road as the reference level).
  - Initial Mechanical Energy: $ME_0 = \frac{1}{2}MV_0^2$
- **Final State (at $x = D$):**
  - Velocity: $V_f = 0$
  - Kinetic Energy: $KE_f = 0$
  - Potential Energy: $GPE_f = 0$
  - Final Mechanical Energy: $ME_f = 0$

2. Calculate the Work Done by Non-Conservative Forces (W_{nc}): The non-conservative forces are the two friction forces, $\vec{f}$ and $\vec{f'}$. They act over a displacement $\vec{D}$.

- Work is $W = \vec{F} \cdot \vec{D} = F \cdot D \cdot \cos(\theta)$.

- The displacement $\vec{D}$ is in the positive x-direction, while $\vec{f}$ and $\vec{f'}$ are in the negative x-direction. The angle θ is $180°$.
- $W_f = f \cdot D \cdot \cos(180°) = -fD$
- $W_{f'} = f' \cdot D \cdot \cos(180°) = -f'D$
- Total Non-Conservative Work: $W_{nc} = W_f + W_{f'} = -fD - f'D = -(f + f')D$

3. Apply the Work-Energy Theorem:

$$\Delta ME = W_{nc}$$

$$ME_f - ME_0 = W_{nc}$$

$$0 - \left(\frac{1}{2}MV_0^2\right) = -(f + f')D$$

$$-\frac{1}{2}MV_0^2 = -(f + f')D$$

Solving for D:

$$D = \frac{-\frac{1}{2}MV_0^2}{-(f + f')}$$

$$D = \frac{MV_0^2}{2(f + f')}$$